

Monetary Policy

6.1 OBJECTIVES OF MONETARY POLICY

Monetary policy is the active control of financial conditions by the central bank. It now generally has the objective of ensuring that the domestic currency has the qualities associated with money, i.e., that it can serve as a unit of account, store of value and medium of exchange. This objective generally requires that the future purchasing power of the currency can be anticipated by the general public and that the technical means of exchanging the currency (banknotes, electronic payment systems) work reliably. Modern monetary policy therefore tends to be geared towards some form of price stability which can be defined in various ways. In the policy mode known as inflation targeting, price stability is defined as a given positive inflation rate (zero inflation is undesirable due to measurement issues and asymmetric risk concerns [9]). In the related price level targeting approach, the central bank defines a target path for the price level and therefore explicitly commits to compensate any past deviations in the future. This commitment is at best implicit in inflation targeting. Over time, both the understanding of price stability, and the best way to express it, have evolved across different central banks.

Some central banks, for instance the US Federal Reserve and Reserve Bank of Australia, operate on a mixed mandate that targets the correlated but distinct aims of price stability and maximum employment.

The existence of a monetary policy target suggests that monetary policy is essentially a control problem, and some central bankers have indeed sought to describe their approach in such terms. The idea of reducing economic problems to the solution of mathematical equations could be linked to the suggestion by Karl Marx that 'a science can only be considered to be fully developed when it reached the stage where it can utilise mathematics'. In practice, the large number of parameters such a model would have to incorporate, while the number of observations is limited by the low frequency of collection of the relevant data, precludes the proper calibration of such an ambitious model. In addition, it may be incorrect to assume that central banks have sufficient tools to act in an optimal way even if they had sufficient information.

One can indeed argue that the existence of a complete model of optimal monetary policy under a rational expectations framework would in fact contradict the idea of active monetary policy. By way of *reductio ad absurdum*, if a complete model existed, it would eventually be known to all economic actors. The central bank would under these



FIGURE 6.1 Example of a Japanese castle wall (Osaka Castle). Source: ALEXANDER DÜRING.

circumstances not have any way in which to influence the decisions of others, having to follow a policy path that is known to everybody else¹. In essence, this would be a world of real business cycle models where the only uncertainty stems from random jumps in productivity.

This raises the question whether the absence of a perfect monetary policy approach is a drawback for the economy. The answer is not immediately obvious and instead depends on one's economic model. Followers of the Minsky theory of the business cycle [58] would argue, for instance, that the shocks generated by occasions when monetary policy decisions later turn out to have been mistaken can flush out elevated leverage before it becomes excessive.

An illustration of this idea can be found in the construction of Japanese castle walls (Figure 6.1). Unlike European castles, the slope of Japanese walls becomes progressively steeper towards the top. The common justification for this design is that it creates an adverse selection process for attacking soldiers attempting to climb these walls. Less capable soldiers are likely to fail close to the bottom and may sustain minor injuries when they fall. The most capable soldiers will not fall off the wall until they reach a height at which their fall will result in major, or even fatal, injuries. The wall design therefore selectively apportions the maximum damage to the most dangerous enemies². In the realm of monetary policy, a central bank may be able to offset incoming shocks

¹In economics, these arguments are known as the Lucas Critique (historical correlations are not a good guide to future reactions to monetary policy) and Goodhart's Law (an indicator becomes non-informative when it is chosen as a policy target).

²An unkind observer with a mind focused on physics might argue instead that the inward slope offsets the pressures generated by a quasi-liquid soil in case of an earthquake, and that the absence of this consideration in Europe explains the special design of Japanese castles. However, a perfectly reasonable alternative explanation is no excuse to dismiss a good story.

to the economy by a well-executed monetary policy path. However, this could leave the economy even more exposed to shocks that are either beyond the capacity of the central bank to react, or could not have been foreseen by the central bank.

The lesson of the Japanese castle wall is that there is a trade-off between the steady state performance of an economy and its resilience to shocks. A central bank that can hold its economy close to a state of maximum output (however defined) may find that it is least able to prevent sharp deteriorations in economic conditions simply because there is more room to fall. In other words, the longer an economy grows beyond potential, the more time there is to build up excessive risk positions.

Put more bluntly, active monetary policy may have a purpose not because it is always right but precisely because it may on occasion be wrong. Authors like Cooper [26] who perceive a conflict between the notion of efficient financial markets on one hand, and the need for a central bank on the other, may therefore be missing two important points. First, the provision of money by central banks needs to be carefully managed to endow this money with the qualities expected of it, namely its role as a store of value. If this non-trivial management is not provided in a reliable fashion, markets cannot work. Second, even if the central bank occasionally makes mistakes in this management, these may not always have negative long-term consequences.

That being said, central banks tend to interpret their mandates as leaning against the business cycle instead of aggravating it³, in other words, as dampening volatility instead of ensuring a certain level of uncertainty. As will be discussed later, the relationship between the proclaimed stance of the central bank, and the amount of uncertainty it allows to occur around it, are an important consideration for monetary policy frameworks.

It should be realised that this 'leaning against the cycle' is a good textbook concept but translates into difficult decisions in real life. A central bank looking to dampen aggregate demand in a higher inflation situation may appear to be trying to manage a macroscopic variable. The economic variable of lower aggregate demand, however, translates into a loss of employment for a macroscopically significant number of workers. Explaining to the affected families that their personal income levels need to be reduced to protect the greater good of price stability is a difficult task, especially because this apparent link suffers from a fallacy of decomposition. The central bank does not terminate individual employment contracts, but merely changes probabilities.

In today's world, this has led to a structural change in how central banks communicate with the public. In the not too recent past, central banks shrouded themselves in an aura of mystique and revealed only a limited amount of information about the motives behind their actions. Famously, the Fed did not even disclose its target rate until February 1994 [33]. When the ECB started regular press conferences after Governing Council meetings, and later tied the meeting schedule to the press conference schedule, this level of transparency was unique⁴. Today, even the Fed has regular press conferences. To date, this increased level of transparency has not hurt the credibility of central banks.

³Former Federal Reserve Chairman William McChesney Martin memorably described this task as 'to take away the punch bowl just as the party gets going'.

⁴Visitors to Frankfurt am Main may notice that the Bundesbank building is located on a hill at some distance from the city centre while the original ECB building is very central. In this case, the visible location choices are symbolic for communication approaches.

6.2 MONETARY POLICY UNDER INFLATION TARGETING

In order to discuss the impact of active monetary policy on inflation, it is instructive to start from the basic quantity equation of the price level:

$$MV = PY \quad (6.1)$$

Here, M is the money stock, V the velocity of money, P the price level, and Y the total volume of traded goods and services. For all practical purposes, Y can be thought of as a proxy of GDP; the only difference between the two is changes to inventories. The equation expresses simply that the total volume of things paid for during a given period (PY) is equal to the money handed over in exchange (MV).

Solving for P and taking the total differential gives:

$$dP = \frac{M}{Y}dV + \frac{V}{Y}dM - \frac{MV}{Y^2}dY \quad (6.2)$$

The price level therefore depends on three control variables in this model: Higher prices can be obtained by higher velocity of money, a larger money stock, or lower output. Note that the total differential makes no assumption about correlation, or indeed causal relationships, between the three control variables.

Economists generally agree only that the term dV is unsuitable for monetary policy because it is impossible to control the way in which money is used in the economy. V can react substantially to changes in investment climate as liquidity preferences change. Beyond this point, economists split into monetarists and Keynesians⁵ according to the logical precedence assigned to the dM and dY terms.

A Keynesian economist views the dY term as central to official guidance of the economy and there are many factors in which this term can be controlled, including monetary policy. An important endogenous determinant of the near-term evolution of dY is the current output gap, i.e., the difference between actual and potential (however defined) output [22]. Because there is an assumption that the output gap is mean-reverting to some degree⁶, monetary policy can use measures of the output gap (unemployment rate, capacity utilisation, investment spending) as clues for monetary policy. On this basis, the policy interest rate set by the central bank is determined by the natural rate of interest for the economy r^* on the one hand, and the distance of inflation from its target on the other. Keynesians regard the dM term as reacting passively to dY . This view is supported by the observation that most of M in the usual definitions ($M2$ and $M3$) is not central bank money, but monies created by commercial banks. In the euro area, only about 10% of $M3$ are currency in circulation and bank reserves ($M0$); the rest are largely deposits. However, changes in M can have effects on the economy even in this framework, if largely through relative asset prices and signalling. The difficulty of this approach is that it relies on unobservable variables, such as the output gap, to judge the appropriate direction for monetary policy.

⁵Apologies to any economists who object to being subsumed into these two groups despite having subtle but important differences with either. Neo-classical economics are subsumed here under the Keynesian school.

⁶Mean-reversion is sometimes implicit in the use of Hodrick-Prescott filters in the estimation of the output gap.

For monetarists, the dM term is paramount for central bankers because the dY term is not subject to central bank control. The output gap is rejected as a monetary policy indicator because it cannot reliably be measured in real time⁷. Monetarists agree that dY has an influence on the price level; the contention is simply that a central bank should not look to intervene in real variables⁸. In this world view, money growth dM is simply given by a desired inflation level dP plus an a term that accounts for the growth in money demand arising from economic growth dY and shifts in money usage (velocity), given by dV . This is the theoretical background behind the money growth target approach that was taken by the Bundesbank. As long as the coefficients of Equation (6.2) are reasonably stable, a central bank that targets money growth is indistinguishable from one that targets inflation [28]. The problem with money growth targeting is that the target quantity must be reasonably wide so as to be less sensitive to short-term fluctuation related to transaction money demand (which tends to be inside $M0$). Even so, there are multiple reasons to hold money other than in order to spend it on goods and services.

All that being said, the left side of Equation (6.1) is somewhat of a fiction. There are multiple definitions of M ($M0$ – $M3$, $M2$ + CD , base money) vying for the attention of central bankers, and V is unobservable. The equation is therefore not much more than a definition of V , rather than a useful guideline to monetary policy.

Brushing aside this technical difficulty, one finds central banks with inflation targets in agreement on a hierarchy of price level control, shown in the chart below:

$$\begin{array}{rcl}
 \text{Quantity (open market operations or lending)} & & \\
 \downarrow \text{(III)} & & \\
 \text{Price (interest rates on central bank money)} & & \\
 \downarrow \text{(II)} & & (6.3) \\
 \text{Quantity (money supply)} & & \\
 \downarrow \text{(I)} & & \\
 \text{Price (goods and services)} & &
 \end{array}$$

Central banks control the price level of consumer goods by adjusting the money supply M , ie., the quantity of money (I). This quantity cannot be controlled directly by the central bank because most of M is not central bank money. Instead, central banks control the demand for M (II) via interest rates (monetarist explanation) or signal their stance on the real economy via interest rates (Keynesian explanation). The ECB refers to (II) as the monetary transmission mechanism. Interest rates are simply a price of money and by changing the price, central banks can look to influence demand for money.

Note that relationships (I) and (II) operate with a long and variable lag. The creation of the money components outside $M0$ require decisions of private actors to borrow, and changes in the availability of money may feed into price setting mechanisms only slowly, given imperfect information and market rigidities. A useful benchmark is a lag of 12–18 months.

⁷This measurement problem explains why the Taylor rule works well as an ex-post explanation of central bank action but much less well as a forecasting tool.

⁸The tradition of monetarism in Germany is part of the explanation of why the court cases filed in that country against ECB monetary policy usually allege that the ECB oversteps the mandate of a central bank by acting in ways that affect real variables, instead of only nominal ones. To a Keynesian, it is absurd to assume that the two can be cleanly separated.

Given that there are two mutually exclusive interpretations of central banking, the European Central Bank had to choose one to guide its own policy decisions when it was set up in 1999. In line with the behaviour of other European institutions in similar situations it adopted both. The two pillars of the ECB policy framework are a monetarist M3 analysis and a potentially Keynesian inflation analysis. While the early chief economists at the ECB used to interpret the second pillar narrowly and use output gap measures mainly as a quality control tool of macro-economic forecasts, more recent communications put more emphasis on the 'slack in the economy' which can be interpreted as Keynesianite references to the output gap⁹.

Under normal circumstances (standard monetary policy), some, but not all, central banks control the relevant interest rates via quantities (III), namely the amount of daily open market operations. The Fed and the Bank of Japan fall into this group; the ECB uses a mix of instruments but under normal circumstances influences interest rates directly via the minimum rate at which it allocates cash at the main refinancing operations.

Demand for money can change due to purely technical factors such as tax and wage payments, holiday gift shopping seasons, etc. At high frequencies money demand is not perfectly responsive to policy interest rates¹⁰.

The alteration between quantity and prices in the control hierarchy Figure 6.3 can cause breakdowns when for some reason one of the relationships no longer works as anticipated. For instance, price levels may become insensitive to money supply when households simply hoard cash. In the framework of Equation (6.2) one could find that:

$$\frac{M}{Y}dV = -\frac{V}{Y}dM \quad (6.4)$$

so that the effect of changing M would simply be offset by an opposing change of V . Another way in which this relationship can break down is when $V \rightarrow 0$ which implies $\partial P/\partial M \rightarrow 0$. This situation arises when the general economy loses faith in the domestic currency and resorts to barter or the use of a substitute currency (dollarisation). Price levels then become independent of the domestic money supply simply because domestic money is no longer in use.

Two specific forms of Goodhart's Law may apply to inflation targeting. The first is that long-term inflation expectations of the public should, if the central bank is credible, converge to the central bank's target. Public expectations of inflation would then no longer be informative for policy makers as to the actual outlook for inflation which would remain subject to various shocks and require monetary policy action. The second is that some policy tools can affect market measures of future inflation such as derivatives linked to future price level changes. In such cases, these market measures can also become non-informative.

⁹Keen readers of the ECB's Introductory Statements will note that the economic analysis comes before the monetary analysis, both of which are then cross-checked.

¹⁰This is essentially why the Friedmanite episode of the Fed ended in failure (inelastic money supply led to volatile market interest rates) and why the Bundesbank operated more as an inflation-targeting central bank despite its professed money supply framework [28].

6.3 CENTRAL BANK OPERATIONAL FRAMEWORKS

Central banks can use various tools to control the supply of liquidity to private banks. Each central bank will typically choose a subset of the available tools according to the specific situation of its domestic banking market. The chosen set of tools and the way they are implemented is called the operational framework of the central bank. Factors such as the structure of the banking system, the size and structure of the relevant securities markets, and the relevance of net foreign trade can all influence the design of the operational framework. Note that the framework itself need not be tied to a particular monetary policy objective. The same objectives might be achieved with various frameworks, and the same framework can be used for different, or changing, monetary policy objectives.

In general, however, there are certain common desiderata for operational frameworks (cf. e.g. [10]), as outlined below.

Parsimony Frameworks should not make use of an excessive number of distinct instruments and instead have a small number of clearly understood mechanisms to achieve standard aims (liquidity provision, liquidity absorption, emergency lending, etc.). A surfeit of instruments is likely to confuse market participants as to the exact monetary policy objective inherent in the choice between two similar instruments. Similarly, a central bank should avoid the simultaneous use of discretionary instruments with opposing monetary effects.

Stability Operational frameworks can and should change over time to adapt to changes in the operating environment. Changes to standard tools should, however, not occur without a well-understood and well-communicated justification. One can therefore observe a certain amount of hysteresis in how central banks interact with the market as the side effects of a useful small adjustment may outweigh the potential benefits. It may be noted that the specific institutional setup of a central bank itself is usually also the result of specific historic circumstances rather than the result of an extended period of careful design [53].

Moderation Central banks must apply instruments in order to achieve the desired effects but should do so with the minimum of disruption to private market interactions. This is not only a question of limited mandates, but has financial stability effects. For instance, it is useful to force private banks to ‘test their name’ in the private market on a regular basis, instead of providing liquidity to them from the central bank, so as to gain insights as to the public opinion about their strength.

A potential drawback of moderation is that it creates a stigma for the use of non-standard liquidity facilities, such as the Fed discount window or the ECB dollar-providing operations. Banks may be reluctant to make use of such facilities when there is a tacit or explicit understanding that doing so is a sign of crisis. Given that risk premia charged by private funds providers are normally higher than what a central bank can provide, avoiding the stigma of a central bank facility will normally mean higher funding costs for the bank in question. Whether this

is in the long-term interest of financial stability is not an easy question to answer and central banks take different stances on this issue.

Separation Instruments in the framework should be used for clearly defined purposes, and these purposes should be clearly delineated. For instance, emergency lending should be conducted in ways other than through regular lending operations such as not to confuse the monetary policy signal of the latter.

The separation principle has been blamed for the slowness of the European Central Bank in adopting non-standard monetary policy measures [47]. Such criticism is probably misplaced because hesitation was a global feature of the use of non-standard measures and may be linked to the dictum of then-chairman Ben Bernanke ‘The problem with quantitative easing is that it works in practice, but not in theory.’ Central bank models of the economy before the economic crisis that started in 2008 generally suffered from a Keynesian disdain for financial matters and did therefore not incorporate transmission channels of monetary policy that are the hallmark of non-standard measures, namely bank equity and investment portfolio allocations.

6.3.1 Symmetric interest rate corridors

Central banks outside crisis situations tend to operate in a setting that can be described as a symmetric corridor system. In this setup, interbank interest rates fluctuate inside a range, and central bank liquidity¹¹ provision is minimal, designed to cover exactly the aggregate needs created by bank notes, reserve requirements and transactional account holdings of central bank cash. Such a system requires a minimum of three tools, namely one to provide the bulk of liquidity, one to absorb any excess and a third to cover any shortfalls. The interest rates on these three tools then describe the centre of the corridor, and its lower and upper bounds.

Liquidity provision can take the form of regular auctions or open market operations. Where auctions are used, the volume supplied must match the banking system demand at the desired interest rate. To judge this demand, all factors of liquidity demand must be forecast, and particularly the demand for transactional liquidity can be subject to large fluctuations.

When multiple liquidity-providing operations exist, their combined quantitative effect needs to be taken into account. In principle, all of the banking system’s *aggregate* liquidity needs should be exactly covered by these operations. Banks should then distribute among themselves on a daily basis this liquidity to cover *individual* requirements which fluctuate more than the aggregate demand.

Participation in the liquidity provision operations is ensured by keeping the banking system structurally short liquidity. Traditionally, this has been achieved through reserve requirements that force banks to borrow central bank funds which

¹¹To recall, the term ‘liquidity’ is here used in the sense of central bank funds provided, generally into current accounts held by commercial banks at the central bank. In the US, this is known as ‘reserves’.

then need to be kept at the central bank¹². In today's market, this requirement is no longer necessary to generate bank demand for central bank liquidities as transactional funds demand has risen.

Standing facilities can then be used to cover situations where liquidity does not match expectations, either in aggregate or for individual banks. Excess liquidity can be absorbed either in a specific facility (like the ECB's Deposit Facility) or remunerated at a lower rate (the Fed's Interest on Excess Reserves). When aggregate or individual needs exceed the amount of liquidity available, this can be covered through a specific lending facility. In the case of the ECB, this is the Marginal Lending Facility and for the Federal Reserve the Discount Window.

The use of standing facilities should be the exception, not the norm. There are two reasons for such usage. The first is unexpected liquidity demand, i.e., a forecast error by the central bank. The risk of such errors occurring rises with the time horizons of liquidity provision, so it is least for daily operations and higher for longer ones. More concerning is a situation where both facilities are used at the same time because that means that at least one bank has a liquidity surplus that it is unwilling to lend to at least one other bank with a liquidity shortfall. The distance between the interest rates on the two facilities should be large enough to discourage such behaviour.

Stigmatisation of lending facilities outside the main operations is a constant topic in money markets. On one hand, banks should trade among each other, and a regular use of standing facilities by any one bank can be sign of weakness. In the US, recourse to the Discount Window has traditionally been seen as negative, although at the time of writing the Federal Reserve is encouraging banks not to view it as such. In a similar way, some of the long-term fund supplying operations of the ECB were initially seen as stigmatised.

It should be noted that because the Fed's Open Markets desk in principle conducts daily operations, there should be less risk of an aggregate liquidity shortfall. Therefore, access to the Discount Window may more readily be seen as a sign of individual, rather than aggregate, circumstances. In addition, the Eurosystem has a designated instrument known as Emergency Liquidity Assistance that is reserved for solvent banks with specific liquidity problems. The clear separation between this instrument and the Marginal Lending facility may serve to de-stigmatise the use of the latter.

During the financial crisis, it became obvious that the euro area interbank market had become less efficient than before. As part of the response, the ECB narrowed the interest rate corridor between the Marginal Lending and Deposit Facilities. In principle, this lowers the cost to the banking system of using the Eurosystem balance sheet as a

¹²Traditionally, these reserves would be in specie, or be payable in specie by the central banks, and serve to ensure that a bank can pay out specie on its liabilities on demand. This is the historic reason why the US market refers to current account holdings at the Fed as reserves while the European market simply refers to them as liquidity. In Japan, they are known as current account holdings.

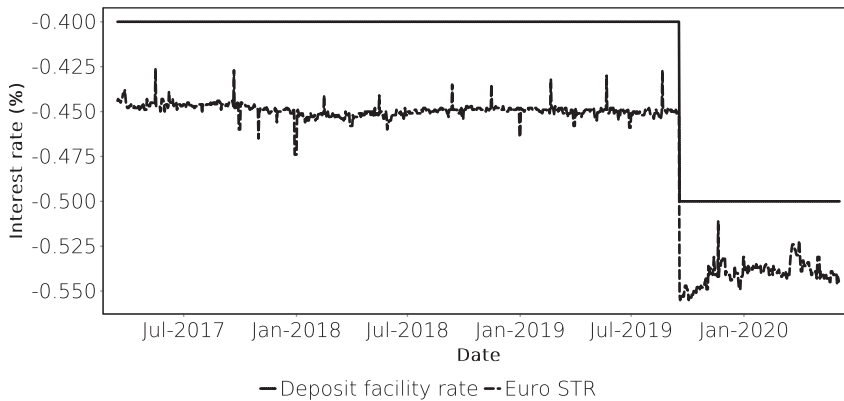


FIGURE 6.2 ECB deposit facility rate and pre-€STR and €STR. Source: From ECB SDW, ECB deposit facility rate and pre-eSTR and eSTR. The information may be obtained free of charge through: <https://sdw.ecb.europa.eu/>. © 2020, European Central Bank.

central counterparty in the money market. At the current juncture, however, there is such a high level of excess liquidity in the system that the corridor is now of lesser relevance, as will be discussed below.

6.3.2 Asymmetric lending corridors

With the financial crisis, and the wider adoption of non-standard monetary policy measures, most central banks' operating frameworks have in principle become asymmetric. In this setting, the market rate of interest does not fall in the middle of a corridor between standing facility rates, but is instead permanently pinned to one side of the corridor. Currently, this is the liquidity-absorbing side, i.e., its lower boundary.

In this setup, the level of liquidity in the banking system is no longer dependent on main refinancing, or standard open market operations. Instead, the banking system operates with a liquidity surplus generated by non-conventional liquidity injections.

One of the side effects of purchase programmes in particular is that central bank reserves were injected in exchange for assets held by non-banks. The non-bank sector therefore had an increasing need to desposit cash with commercial banks which lowered the rates at which such funds traded outside the banking system. This is visible, for instance, in the euro short-term rate (€STR) which includes non-bank transactions with banks and has been consistently below the ECB's deposit facility rate since it has been produced (cf. Figure 6.2). In the US, the Federal Reserve has introduced a new facility, the overnight reverse repo (ON RRP) to absorb liquidity from a wider range of counterparties in order to prevent market rates from falling below the target rate.